Localization & State Estimation

Prepared By

Dr. Islam Zakaria

Lecturer at the Information Technology Department, FCAIH, Capital University

Learning Objectives

- Understand why localization is the hardest problem in robotics
- Differentiate between Local (Tracking) and Global (Kidnapped Robot) localization
- Master the mathematical framework: Bayes Filter and its implementations
- Implement the three pillars: Histogram Filter, Kalman Filter, and Particle Filter
- Analyze the trade-offs: Computational cost vs. Accuracy vs. Robustness
- Recognize why "perfect localization" is impossible (and why that's OK)
- Bridge theory to practice: From math to running code
- **CS Philosophy:** A robot that doesn't know where it is cannot decide where to go. State estimation is not about finding truth—it's about managing uncertainty.

The Localization Problem

- **What Does “Where Am I?” Mean?**
- **In mobile robotics, localization means:**
  - Estimating the robot’s position inside a known environment despite uncertainty.
- This is a **probabilistic estimation problem**, not a deterministic one.

The Localization Problem (Cont.)

Formal Problem Definition

Given:

- **Control Inputs** $u_{1:t}$

  The sequence of motion commands from time **1** to time t:

$$u_{1:t} = \{u_1, u_2, \dots, u_t\}$$

- **Examples:**
  - u_1: Move forward 1m
  - u_2: Rotate 90°
  - u_3: Accelerate

These represent what we commanded the robot to do.

The Localization Problem (Cont.)

- **Sensor Measurements** $z_{1:t}$

  The sequence of observations from time **1** to time **t**:

$$z_{1:t} = \{z_1, z_2, \dots, z_t\}$$

- **Examples:**
  - Z_1: "I see a door"
  - Z_2:"Wall detected at 2.1m"
  - Z_3:"Landmark detected"

  These represent what the robot observed.

The Localization Problem (Cont.)

- **Map** m

  A known representation of the environment:

  - **Occupancy grid**
  - **Landmark map**
  - **Feature-based map**

  The map provides prior structural knowledge.

The Localization Problem (Cont.)

❑ What Do We Compute?

$$Bel(x_t) = p(x_t | z_{1:t} u_{1:t} m)$$

Where:

- x_t: Robot position at time t
- $Bel(x_t)$: Belief (probability distribution) over position
- $p(\cdot)$: Probability
- $|$: Given

The Localization Problem (Cont.)

Why Is Localization Hard?

Localization is difficult because real systems are uncertain.

Challenge	Reality
Sensors are noisy	Measured 5.0m → Actually 5.0 ± 0.2m
Motion is imperfect	Commanded 1m → Actually 0.95m
Environment changes	Door in map → Now open/closed
Symmetry problem	Identical corridors → Multiple possible positions

The Localization Problem (Cont.)

❑ Nature of the Problem

Localization is a:

- Sequential estimation problem
  - Updating a probability when new evidence arrives.
- Uncertainty propagation problem

This formulation is fundamental in probabilistic robotics.

The Localization Problem (Cont.)

- **Conceptual Insight**

  Think of localization as maintaining a **heatmap over the map**:

  - **Bright areas** → **High probability**
  - **Dark areas** → **Low probability**

  As new data arrives:

  - Motion updates shift the distribution
  - Sensor updates reshape it

- **The Golden Rule of Localization**

  You cannot know exactly where you are. You can only know the probability that you are somewhere.

Two Types of Localization

- Localization problems are **not** all equally difficult.
- There are **two** major categories depending on initial uncertainty.
  - **Position Tracking (The Easier Case)**
  - **Global Localization (The Hard Case)**

Position Tracking

The robot **approximately knows** its initial position and only needs to maintain and refine that estimate over time.

Example:

A cleaning robot starting from its charging dock (e.g., **Roomba**).

Characteristics

- Initial uncertainty is **small**
- Probability distribution is **unimodal**
- Belief resembles a **single Gaussian (bell-shaped curve)**
- The robot tracks small motion and sensor errors

Position Tracking (Cont.)

- **The Core Problem**
  - Prevent the uncertainty from growing too large or drifting away from reality.
  - Since motion adds noise and sensors are imperfect, the belief spreads over time.
    - It means that the uncertainty about the robot's position increases as it moves.
- **Mathematical Nature**
  - If noise is approximately Gaussian, the belief can be modeled as:

$$Bel(x_t) \sim \mathcal{N}(\mu_t, \Sigma_t)$$

Where:

- $\mathcal{N} \rightarrow$ Normal (Gaussian) Distribution
- $\mu_t \rightarrow$ Estimated position
- $\Sigma_t \rightarrow$ Uncertainty (covariance)

Global Localization – The "Hard Case"

- **The Core problem**

  Global Localization occurs when the robot must determine its position without a known starting point.

  - **The Problem:** The robot has a map but **zero information** about its initial coordinates or orientation.
  - **The "Kidnapped Robot" Problem:** A specific sub-case where a robot is physically moved to a new location while powered off or its sensors are blocked, forcing it to recover its position from scratch.

- **Key Characteristics**

  - **High Initial Entropy:** Uncertainty is at its maximum.
  - **Multimodal Uncertainty:** Unlike simple tracking where the belief is a single “peak" (Gaussian), the robot may believe it is in several different, equally likely places simultaneously (e.g., "I am either in Corridor A or Corridor C").

Global Localization vs. Position Tracking

Feature	Position Tracking	Global Localization
Initial State	Known	Unknown
Belief Shape	Unimodal (One peak)	Multimodal (Many peaks)
Complexity	Low (Local search)	High (Global search)
Typical Tool	Kalman Filter	Particle / Histogram Filter

The Bayesian Framework

❑ The Recursive Estimation Cycle

- Localization is **not** a one-time calculation; it is a continuous loop.
- The robot constantly refines its belief as it moves and perceives the world.
- This loop is known as the **Recursive Bayes Filter**.

The Bayesian Framework (Cont.)

1. Prediction (Action Update)

"Based on where I was and what I did, where might I be now?“

$$\overline{Bel}(x_t) = \int p(x_t|u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

- **u_t (Control Input):** The command given to the robot (e.g., "move forward 1 meter").
- **$p(x_t|u_t,x_{t-1})$ (Motion Model):** The probability of being at **x_t** given the previous state and the action taken. This accounts for mechanical "drift" or wheel slip.
- $\overline{Bel}(x_t)$: **This is the Prior belief**. It represents the robot's guess after moving **but before looking at its sensors**.
- **Note: we are summing up (integrate) all possible previous positions (x_{t-1}) to find the most likely new position.**

The Bayesian Framework (Cont.)

Example: Imagine a robot in a long corridor.

- **Before moving:** Its belief Bel(x_{t-1}) is a sharp peak at the 5-meter mark.
- **The Command:** "Move 1 meter forward“.
- **The Reality:** The wheels might slip, or they might grip perfectly. The robot might end up at 5.9m, 6.0m, or 6.1m.
- **The Integral's Job:** It looks at the 5-meter mark (and the 4.9m and 5.1m marks, just in case) and calculates the likelihood of reaching 6.0m from **all** of them.
- The result is a new distribution $\overline{\mathrm{Bel}}(x_t)$ that is centered around 6.0m but is **wider and shorter** than the original. **The integral is what physically "spreads" the uncertainty**.

The Bayesian Framework (Cont.)

2. Correction (Measurement Update)

"Based on what I see, how probable is each possible position?".

$$Bel(x_t) = \eta\, p(z_t|x_t)\, \overline{Bel}(x_t)$$

- **z_t (Measurement):** The data coming from sensors (Lidar, Camera, Ultrasonic).
- **$p(z_t \mid x_t)$ (Sensor Model):** The likelihood of seeing measurement **z_t** if the robot were actually at position **x_t** .
- **η (Eta):** A normalization constant. It ensures that the final probabilities all sum up to **1**.
- **Bel(x_t): This is the Posterior belief**. It is the refined, updated "**truth**" the robot uses for its next step.

The Bayesian Framework (Cont.)

Summary of the Loop

1. **Start** with a belief **$\mathbf{Bel(x_{t-1})}$**.
2. **Act:** Move the robot. The belief "blurs" out and becomes less certain $\overline{\mathbf{Bel}}(\mathbf{x_t})$.
3. **See:** Take a sensor reading.
4. **Update:** Combine the movement guess with the sensor data to "sharpen" the belief $Bel(x_t)$.
5. **Repeat**.

Three Approaches to State Estimation

The Implementation Triad

Filter	Representation	Best For	Complexity
Kalman Filter	Gaussian (mean + covariance)	Tracking, linear systems	$O(n^3)$
Histogram Filter	Discrete grid	Small state spaces	O(grid cells)
Particle Filter	Set of samples	Global localization, non-linear	O(population)

Key Insight: There is NO free lunch. You must choose:

- **Accuracy** → Particle Filter (many particles)
- **Speed** → Kalman Filter (if applicable)
- **Simplicity** → Histogram Filter (if state space small)

Any Questions